

Embedded Control Systems Answer Key

3. a) Communication Software and TCP/IP Protocol Suite (10 marks)

Communication software enables data exchange between computer systems over a network. It ensures reliable, organized, and structured communication using protocol stacks. The most widely used protocol suite is TCP/IP, developed during the early research phase of the Internet. This research introduced the important concept of Packet Switching, where data is divided into small packets and transmitted independently across networks.

Layers of TCP/IP Protocol Suite

- 1 The TCP/IP protocol suite consists of five layers:
- 2 Physical Layer
- 3 Data Link Layer
- 4 Internet Protocol (IP) Layer
- 5 Transport Layer
- 6 Application Layer

1. Physical Layer

The Physical layer defines the electrical and mechanical characteristics of data transmission. It specifies data rate, signal encoding scheme, voltage levels, and transmission medium. It converts bits into signals and sends them over the communication channel.

2. Data Link Layer

The Data Link layer manages link establishment, data transfer from upper layers, and link disconnection. It ensures error detection and frame synchronization. In LAN networks, it is divided into two sublayers: Medium Access Control (MAC) and Logical Link Control (LLC).

3. Internet Protocol (IP) Layer

The IP layer is responsible for logical addressing and routing of packets across networks. It determines the best path for data transmission. Each device connected to the Internet is assigned a unique IP address. IP version 4 uses 32-bit addressing, while IP version 6 uses 128-bit addressing. Currently, IPv4 is widely used but gradually being replaced by IPv6.

4. Transport Layer

The Transport layer provides end-to-end communication between two systems. Since the IP layer does not guarantee reliability, the Transport layer ensures reliable data transfer using acknowledgements, retransmissions, sequencing, and flow control. It includes TCP (Transmission Control Protocol) for reliable communication and UDP (User Datagram Protocol) for fast but unreliable communication.

5. Application Layer

The Application layer acts as an interface between user applications and the network. Two processes on two different end systems communicate through this layer. The application generates a byte stream which is divided into TCP segments, encapsulated into IP datagrams, then into data link frames, and finally transmitted through the physical layer.

Application Layer Protocols

- SMTP – Used for sending electronic mail.
- MIME – Used for sending multimedia content in emails.
- FTP – Used for file transfer between systems.
- TELNET – Used for remote login.
- HTTP – Used for World Wide Web services.
- SNMP – Used for network management (runs over UDP).

TCP/IP in Embedded Systems

To make an embedded system network-enabled, the TCP/IP protocol stack is integrated with the operating system and application software. The complete software stack is then loaded into the memory of the embedded system, enabling communication over the Internet.

Thus, the TCP/IP protocol suite plays a vital role in modern communication software by providing reliable, scalable, and standardized network communication.

3.b) Role of Bus Structure (10 Marks Answer)

The **bus structure** is a communication system that connects the major components of a computer system such as the CPU, memory, and input/output devices. It provides a shared pathway for data transfer, reducing the need for multiple direct connections between components. In embedded control systems and microprocessor-based systems (such as Intel 8086), the bus structure plays a crucial role in overall system operation.

The primary role of the bus structure is to enable communication between:

- CPU
- Main Memory
- I/O Devices

Instead of separate wiring between each component, a common bus is used, which simplifies system design and improves efficiency.

1. Address Bus

The **Address Bus** carries the address of the memory location that the CPU wants to access.

- It is **unidirectional** (CPU → Memory/I/O).
- Determines memory capacity.
- Example:
In Intel 8086, 20 address lines allow addressing up to **1 MB memory** (2^{20}).
- **Role:** Identifies *where* data should be read from or written to

2. Data Bus

The **Data Bus** carries actual data and instructions.

- It is **bidirectional**.
- Bus width determines how much data can be transferred at once.
- In Intel 8086, the data bus is 16-bit wide.

Role: Transfers the actual information between CPU, memory, and I/O devices.

3. Control Bus

The **Control Bus** carries control and timing signals such as:

- Read (RD)
- Write (WR)
- Memory/I/O select
- Interrupt signals

Role:

- Specifies *what operation* to perform (read/write).
- Ensures proper timing and coordination.
- Prevents data conflicts.

4. Bus Architecture

- Bus architecture defines **how buses are organized in a computer system**.
- It determines how CPU, memory, and I/O devices communicate.
- It affects; System performance, Scalability, Cost, Complexity

Types of Bus Architecture-

- a. Single Bus Architecture
- b. Hierarchical Bus Architecture

4.1 Single Bus Architecture

- All components share one common bus- CPU, Memory, and I/O devices use the same communication path.
- Only one device can use the bus at a time.
- Working Principle-
 - a. CPU places address on bus.
 - b. Control signals are sent.
 - c. Data is transferred.
 - d. Other devices must wait until bus becomes free
- Pros- Low cost, simple design, easy to implement
- Cons-Bus bottleneck, Limited performance, Poor scalability

4.2 Hierarchical Bus Architecture

- It uses multiple buses arranged in levels which separates high-speed and low-speed communication.
- Reduces traffic on main bus.
- Working Principle-
 - a. CPU communicates with memory via high-speed bus.
 - b. I/O devices use a separate bus.
 - c. Bridges connect different buses.
 - d. Multiple transfers can occur efficiently.
- Pros-Higher performance, Reduced bottleneck, better scalability, Supports multiple devices
- Cons-More complex design, Higher cost, Requires bus controllers/bridges

5. Bus Master, Slaves, Arbitration

5.1 Bus Master

- In Bus System, Devices that control the Bus is called as Bus Master
- Only one master can control the bus at a time.
- Bus Master Initiates read/write operations, Places address on address bus, Generates control signals & Controls data transfer.
- E.g. CPU, DMA Controller & Bus Controller

5.2 Bus Slaves

- In Bus System, Devices that responds to bus master is called as Bus slaves
- Bus Slaves respond to master request, Sends or receives data, But Cannot initiate communication
- E.g. RAM, ROM, I/O Devices & Peripheral chips

5.3 Bus Arbitration

- Bus arbitration is the process of Selecting one master among multiple requesters to control the bus.
- If multiple masters try to use the bus simultaneously can cause Data corruption & System instability.
- It ensures fair access, no conflict & organized communication
- Types of Bus Arbitration-
 - a) Centralized Arbitration- A single bus arbiter performs the required arbitration
 - b) Distributed Arbitration- All devices participating in the selection of the next bus master

